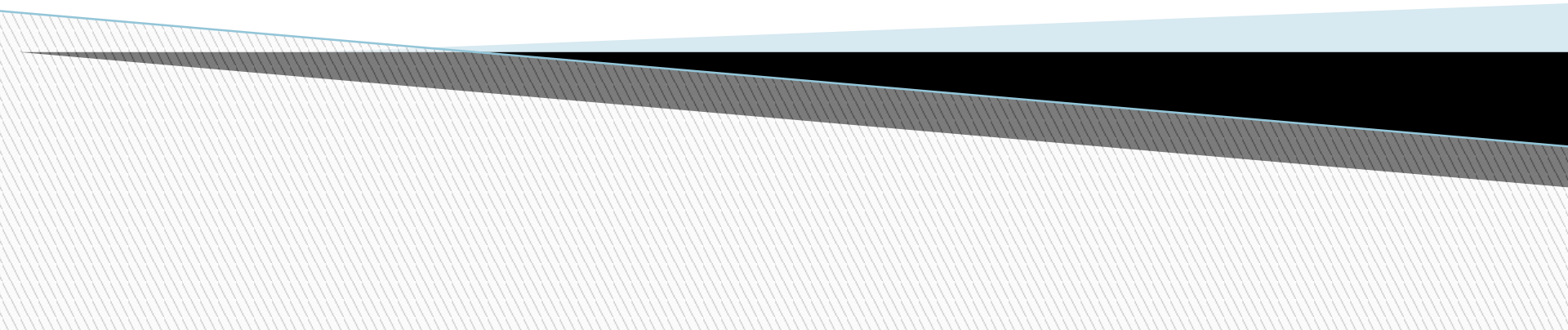


Open Economy

Week 12



OPEN AND CLOSED ECONOMIES

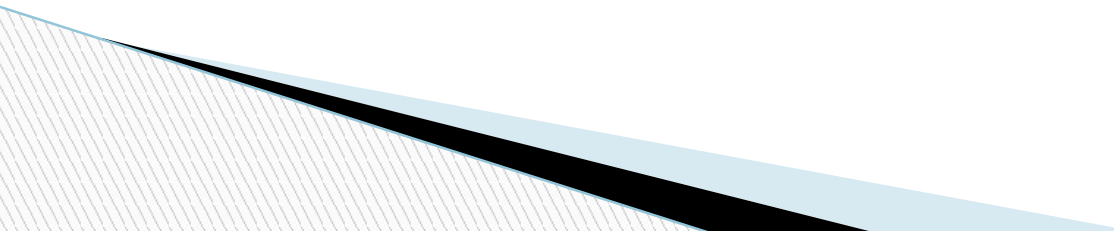
- ❑ A *closed economy* does not interact with other economies in the world.
 - There are no exports, no imports, and no capital flows.
- ❑ An *open economy* interacts freely with other economies around the world in two ways:
 - It buys and sells goods and services in world product markets.
 - It buys and sells capital assets in world financial markets.

How to measure The Degree of Openness?

- ❑ Ratio of exports or imports to GDP over time
- ❑ The value of total trade (imports plus exports) as a percentage of GDP called Trade Dependence Index (TDI)

$$\text{expor+impor} / \text{GDP}$$

In this topic, we will discuss about....

- ❑ The flow of goods: exports, imports, net exports
 - ❑ The flow of financial resources: net capital outflow
 - ❑ The prices for international transactions: real and nominal exchange rates
 - ❑ The Theory of exchange-rate determination: purchasing-power parity
 - ❑ Supply and demand for loanable funds and for foreign-currency exchange
- 

THE FLOW OF GOODS: Exports, Imports, Net Exports

- ❑ *Exports* are goods and services that are produced domestically and sold abroad.
- ❑ *Imports* are goods and services that are produced abroad and sold domestically.
- ❑ *Net exports* (NX) are the value of a nation's exports minus the value of its imports.
 - Net exports are also called the *trade balance*.

Important to note:

How nations record international accounts?

- See Samuelson & Nordhaus (2010) ch.27
p.545-547

Factors that affect exports, imports and net exports

- ❑ The **tastes** of consumers for domestic and foreign goods.
 - ❑ The **prices of goods** at home and abroad.
 - ❑ The **exchange rates** at which people can use domestic currency to buy foreign currencies.
 - ❑ The **incomes of consumers** at home and abroad.
 - ❑ The **costs of transporting goods** from country to country.
 - ❑ The **policies** of the government toward international trade.
- 

Output Determination with Foreign Trade

GDP	Domestic Demand (C+I+G)	X	M	$NX=X-M$	Total Spending (C+I+G+NX)	Tendency of Economy
4.100	4.000	250	410	-160	3.840	Contraction
3.800	3.800	250	380	-130	3.670	Contraction
3.500	3.600	250	350	-100	3.500	Equilibrium
3.200	3.400	250	320	-70	3.330	Expansion
2.900	3.200	250	290	-40	3.160	Expansion

Output Determination with Foreign Trade

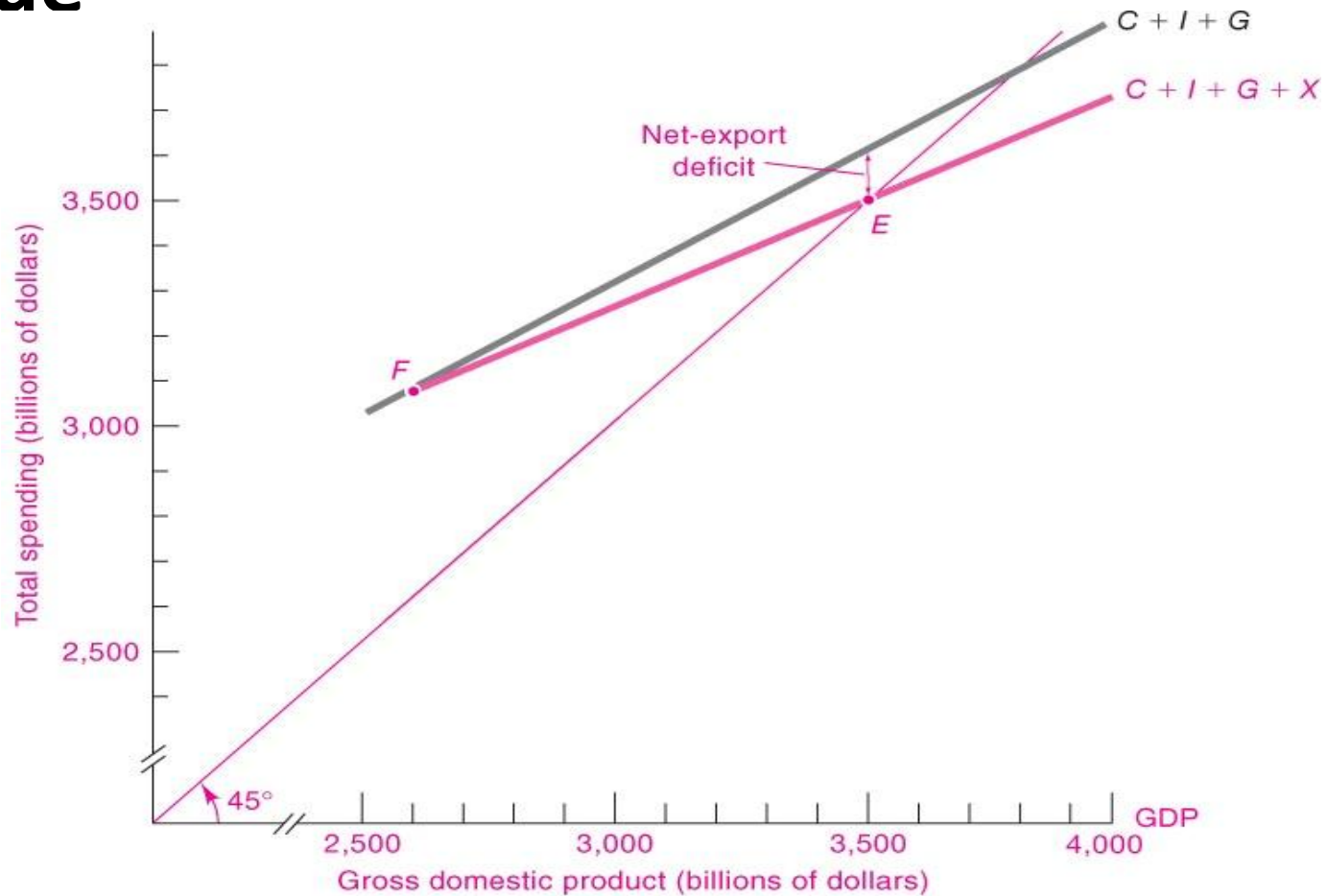


FIGURE 28-2. Adding Net Exports to Domestic Demand Gives Equilibrium GDP in the Open Economy

Short-Run Impact of Trade on GDP

There are two major new macroeconomic elements in the presence of international trade

1. We have a fourth component of spending, the net export (NX).
2. An open economy has different multipliers for private investment and government domestic spending because some spending leaks out to the rest of the world.

The Open-Economy Multiplier

- ▶ The open economy multiplier shows how, in the short run when there are unemployed resources, changes in trade will effect aggregate demand, output, and employment

$$\text{Open economy multiplier} = \frac{1}{MPS + MPM}$$

where MPS = marginal propensity to save and MPM = marginal propensity to import.

- ▶ Because a fraction of any income increase leaks into imports in an open economy, the open multiplier is smaller than the multiplier for a closed economy

THE FLOW OF FINANCIAL RESOURCES: Net Capital Outflow

- NCO is the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreigner
- NCO is the net flow of funds being invested abroad by a country during a certain period of time (usually a year).
 - A positive NCO means that the country invests outside more than the world invests in it;
 - a negative one, that the world invests in the country more than the country invests in the world.

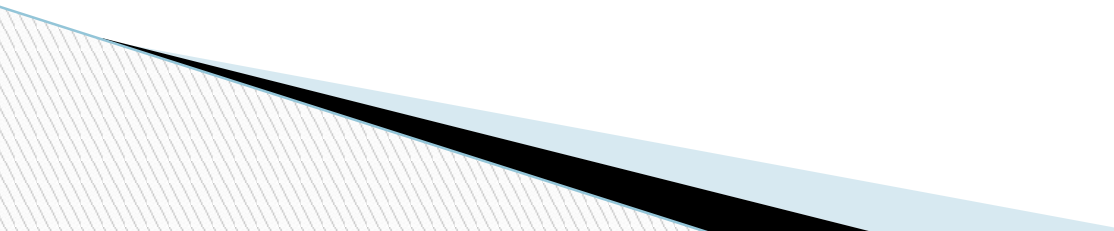
THE FLOW OF FINANCIAL RESOURCES: Net Capital Outflow

- ▶ NCO is the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreigner.
- ▶ Nco adalah pembelian aset luar negeri oleh penduduk dalam negeri dikurangi pembelian aset dalam negeri oleh asing
- ▶ NCO is the net flow of funds being invested abroad by a country during a certain period of time (usually a year).
- ▶ Jadi nco ini aliran dana bersih yang diinvestasikan ke luar negeri oleh suatu negara selama jangka waktu tertentu
 - A positive NCO means that the country invests outside more than the world invests in it;
 - + invest keluar negeri > invest ke dalam negeri
 - a negative one, that the world invests in the country more than the country invests in the world.

THE FLOW OF FINANCIAL RESOURCES: Net Capital Outflow

- Flow of capital takes two forms:
 - Foreign direct investment
 - Foreign portfolio investment

Variables that Influence *Net Capital Outflow*

- ❑ The real interest rates being paid on foreign assets.
 - ❑ The real interest rates being paid on domestic assets.
 - ❑ The perceived economic and political risks of holding assets abroad.
 - ❑ The government policies that affect foreign ownership of domestic assets.
- 

The Equality of Net Exports and Net Capital Outflow

- Net exports (NX) and net capital outflow (NCO) are closely linked.
- For an economy as a whole, NX and NCO must balance each other so that:

$$NCO = NX$$

- This holds true because every transaction that affects one side must also affect the other side by the same amount.

The Equality of Net Exports and Net Capital Outflow

- When $NX > 0$ (trade surplus)
 - Selling more goods and services to foreigners than it is buying from them
 - From net sale of goods and services
 - Receives foreign currency
 - Buy foreign assets
 - Capital - flowing out of the country: $NCO > 0$

The Equality of Net Exports and Net Capital Outflow

- When $NX < 0$ (trade deficit)
 - Buying more goods and services from foreigners than it is selling to them
 - The net purchase of goods and services
 - Needs financing
 - Selling assets abroad
 - Capital - flowing into the country: $NCO < 0$

The Equality of Net Exports and Net Capital Outflow

- In order **to pay for the exports**, the country purchasing the exports will **trade assets to obtain currency** to make those payments. The value of those payments is equal to the net exports. Net capital outflow is what the country selling the exports provided in currency to obtain the assets traded for the currency. You can't get the money to buy the export goods without first getting the money by selling either currency of your own or another country, or assets for the currency you need to buy the exports
- The value of exports (bananas, ice cream, clothing) produced in country A is always matched by the value of reciprocal payments of some asset (cash, stocks, real estate) made by buyers in other countries to the producers in country A. This value is also equal to the total amount of A's currency traded in the foreign exchange market over that year, because essentially the buyers in other countries trade in their assets (e.g. foreign currency) to convert to equivalent amount in A's currency, and use this amount to pay for A's export products

Saving and Investment in the Open Economy

- Net exports is a component of GDP:

$$Y = C + I + G + NX$$

- National saving is the income of the nation that is left after paying for current consumption and government purchases:

$$Y - C - G = I + NX$$

- Recall that national saving is sum of private saving, $(Y - T - C)$, and public saving, $(T - G)$.

Saving and Investment in the Open Economy

- ▣ National saving (S) equals $Y - C - G$, so:

$$S = I + NX$$

- ▣ For an economy as a whole, NX and NCO must balance each other so that:

$$S = I + NCO$$

THE PRICES FOR INTERNATIONAL TRANSACTIONS: nominal and real exchange rates

- ❑ International transactions are influenced by international prices.
- ❑ The two most important international prices are the *nominal exchange rate* and the *real exchange rate*.

Nominal Exchange Rates

- ❑ The *nominal exchange rate* is the rate at which a person can trade the currency of one country for the currency of another.
- ❑ The nominal exchange rate is expressed in two ways:
 - in units of foreign currency per one U.S. dollar.
 - in units of U.S. dollars per one unit of the foreign currency.

Nominal Exchange Rates

- Assume the exchange rate between the Japanese yen and U.S. dollar is 80 yen to one dollar.
 - One U.S. dollar trades for 80 yen.
 - One yen trades for $1/80$ ($= 0.0125$) of a dollar.

Real Exchange Rates

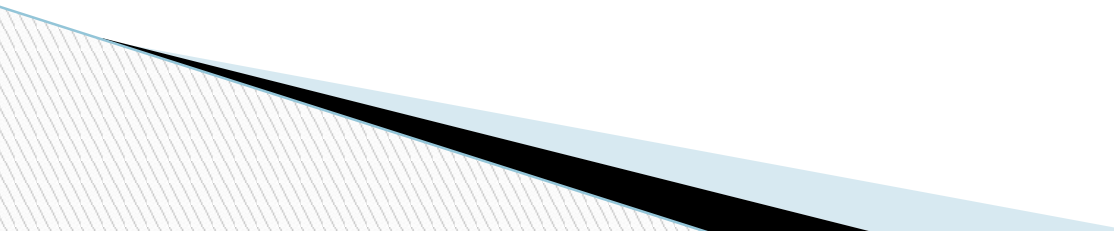
- ❑ The *real exchange rate* is the rate at which a person can trade the goods and services of one country for the goods and services of another.
- ❑ The real exchange rate compares the prices of domestic goods and foreign goods in the domestic economy.
 - If a case of German beer is twice as expensive as American beer, the real exchange rate is $1/2$ case of German beer per case of American beer.

Real Exchange Rates

- The real exchange rate depends on the nominal exchange rate and the prices of goods in the two countries measured in local currencies.

$$\text{Real Exchange Rate } (\varepsilon) = \text{Nominal Exchange Rate } (e) \times \frac{\text{Domestic Price } (P_d)}{\text{Foreign Price } (P_f)}$$

Exchange Rates Depreciation and Appreciation

- ❑ *Appreciation* refers to an increase in the value of a currency as measured by the amount of foreign currency it can buy.
 - ❑ *Depreciation* refers to a decrease in the value of a currency as measured by the amount of foreign currency it can buy.
- 

Exchange Rates Depreciation and Appreciation

- A **depreciation** (fall) in the U.S. exchange rate means that U.S. goods have become cheaper relative to foreign goods.
 - This encourages consumers abroad to buy more U.S. goods, so U.S. **net export rise**.
- Conversely, **an appreciation** in the U.S. real exchange rate means that U.S. goods have become more expensive compared to foreign goods, so U.S. **net exports fall**.

THEORY OF EXCHANGE-RATE DETERMINATION: Purchasing-Power Parity

- Purchasing power = value of money in terms of quantity of goods it can buy
- Parity = equality
- Purchasing-power parity is a theory of exchange rates whereby **a unit of any given currency should be able to buy the same quantity of goods in all countries.**
- The theory of purchasing-power parity is based on a principle called *the law of one price*.
 - According to the law of one price, a good must sell for the same price in all locations.

Implications of Purchasing-Power Parity

- ❑ If the purchasing power of the dollar is always the same at home and abroad, then the exchange rate cannot change.
- ❑ The nominal exchange rate between the currencies of two countries must reflect the different price levels in those countries.

$$e = P_f / P_d$$

Where, P_f = foreign price and P_d = domestic price

What do the prices of a BigMac Hamburger around the world tell us about exchange rates?

■ An example:

- Suppose that price of BMH in Indonesia (july 2017) = \$2.4 (or in rupiah = Rp32126). Then, actual rate of dollar ER should be = $32126/2.4 = \text{Rp}13369.5 \cong \text{Rp}13370$ per US dollar
- Price of BMH in US = \$5.3. Then, dollar ER based on PPP = $32126/5.3 = \text{Rp}6062$ per US dollar (only 45.3% from the actual rate)
 - meaning that Rupiah is undervalued by 54.3% against the US dollar.

■ The Big Mac index (The Economist, 2017)

<http://www.economist.com/content/big-mac-index>

Implications of Purchasing-Power Parity

- When the central bank prints large quantities of money, the money loses value both in terms of the goods and services it can buy and in terms of the amount of other currencies it can buy.

Limitations of Purchasing-Power Parity

- Many goods are not easily traded or shipped from one country to another.
- Tradable goods are not always perfect substitutes when they are produced in different countries.

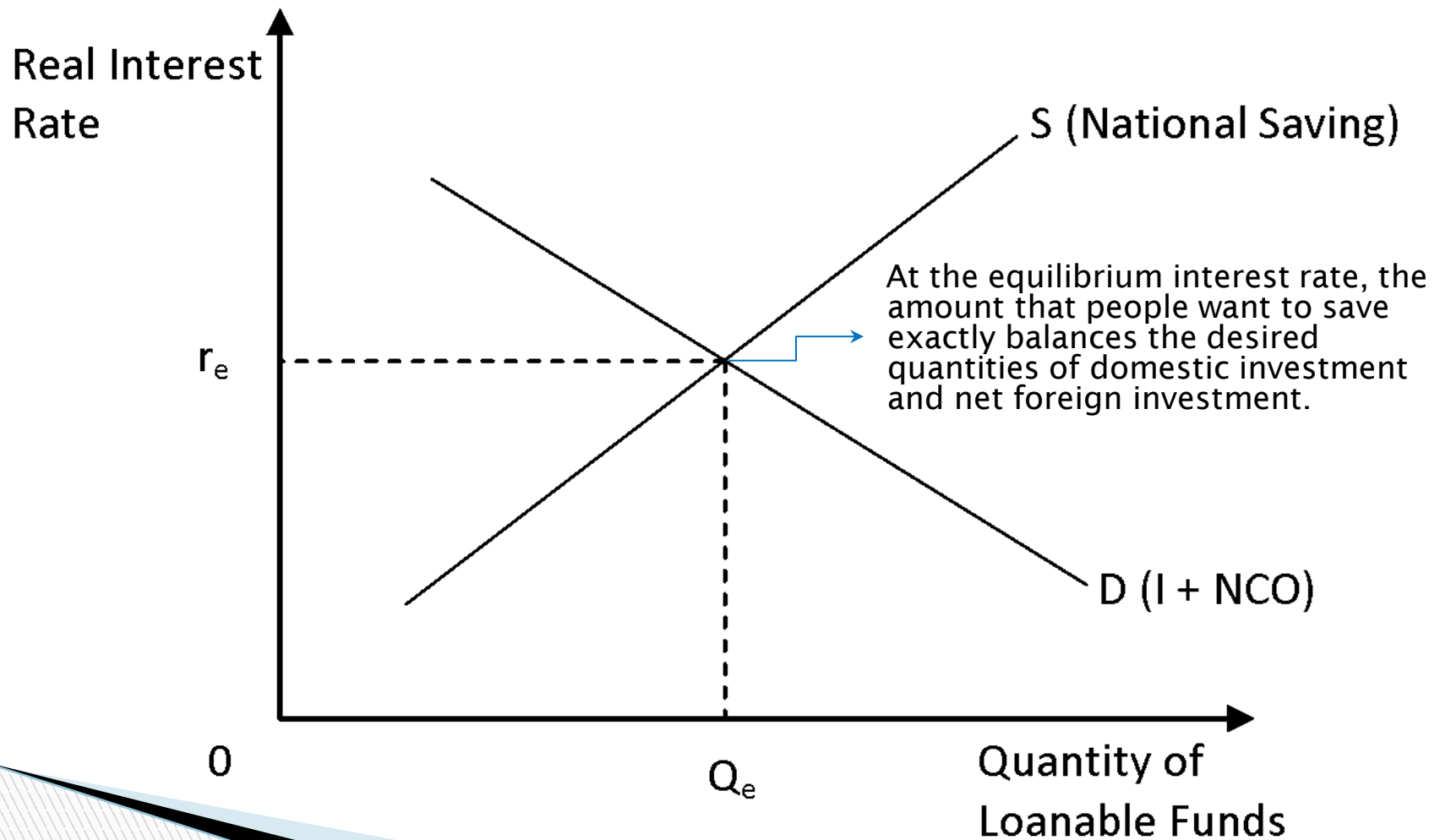
SUPPLY AND DEMAND FOR LOANABLE FUNDS AND FOR FOREIGN-CURRENCY EXCHANGE

- The Market for Loanable Funds (see saving & investment in the open economy at previous slide)

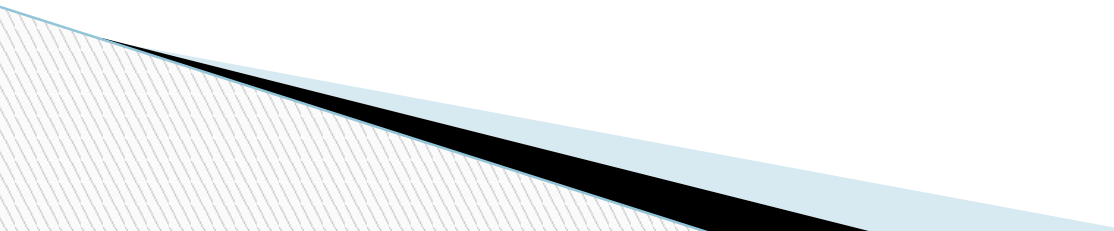
$$S = I + NCO$$

- The supply of loanable funds comes from national saving (S).
- The demand for loanable funds comes from domestic investment (I) and net capital outflows (NCO).

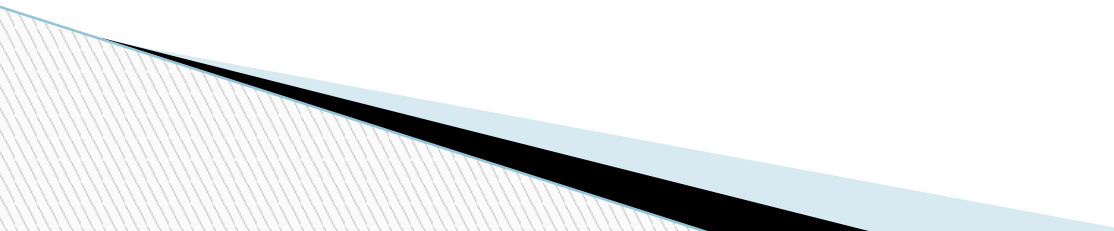
The Market for Loanable Funds



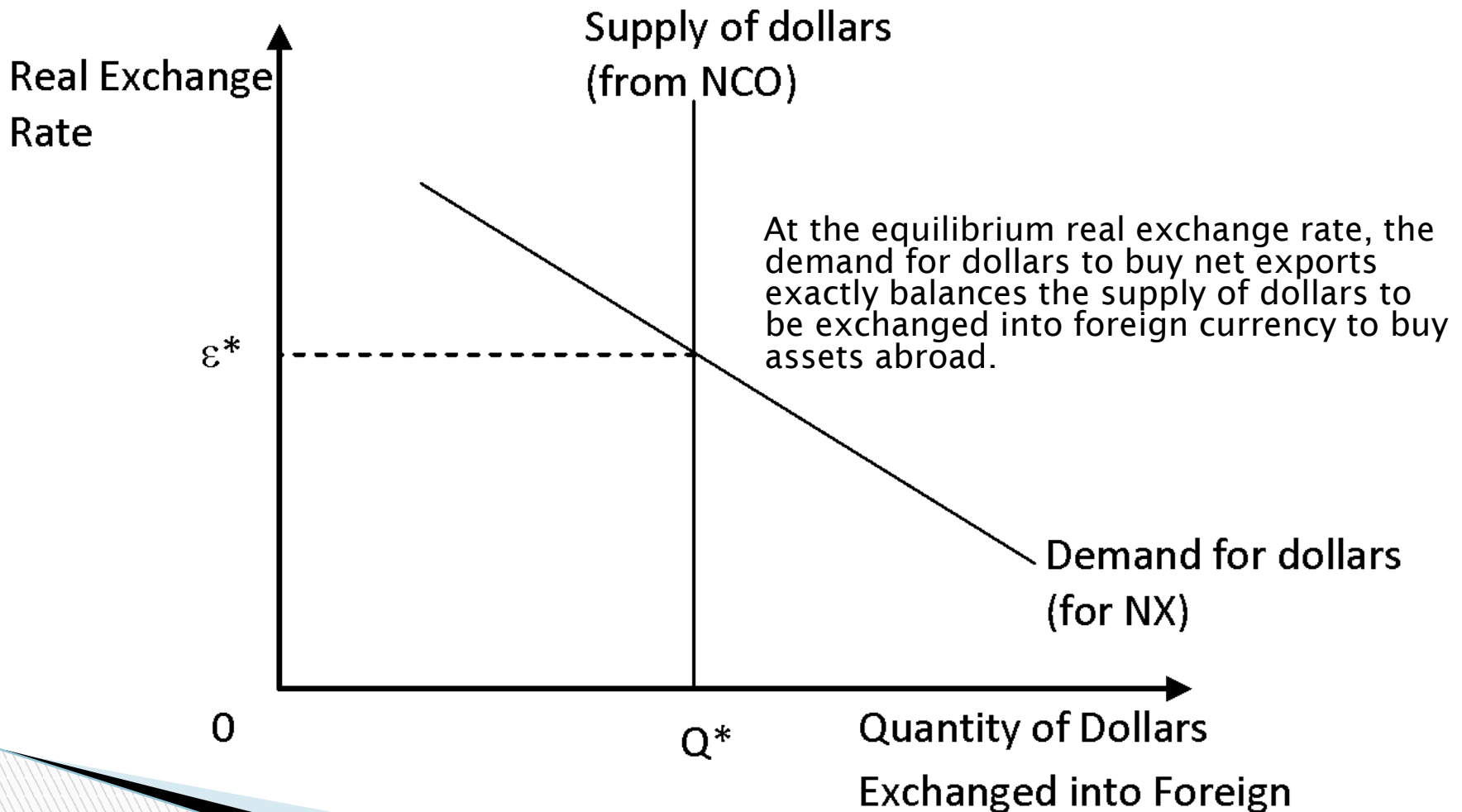
The Market for Foreign-Currency Exchange

- The two sides of the foreign-currency exchange market are represented by NCO and NX .
 - NCO represents the imbalance between the purchases and sales of capital assets.
 - NX represents the imbalance between exports and imports of goods and services.
- 

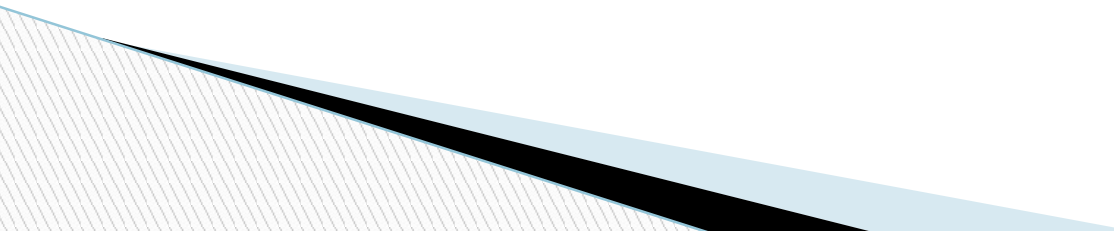
The Market for Foreign-Currency Exchange

- ❑ The demand curve for foreign currency is downward sloping because a higher exchange rate makes domestic goods more expensive.
 - ❑ The supply curve is vertical because the quantity of dollars supplied for net capital outflow is unrelated to the real exchange rate.
 - ❑ The price that balances the supply and demand for foreign-currency is the real exchange rate.
- 

The Market for Foreign-Currency Exchange



The Real Equilibrium in an Open Economy

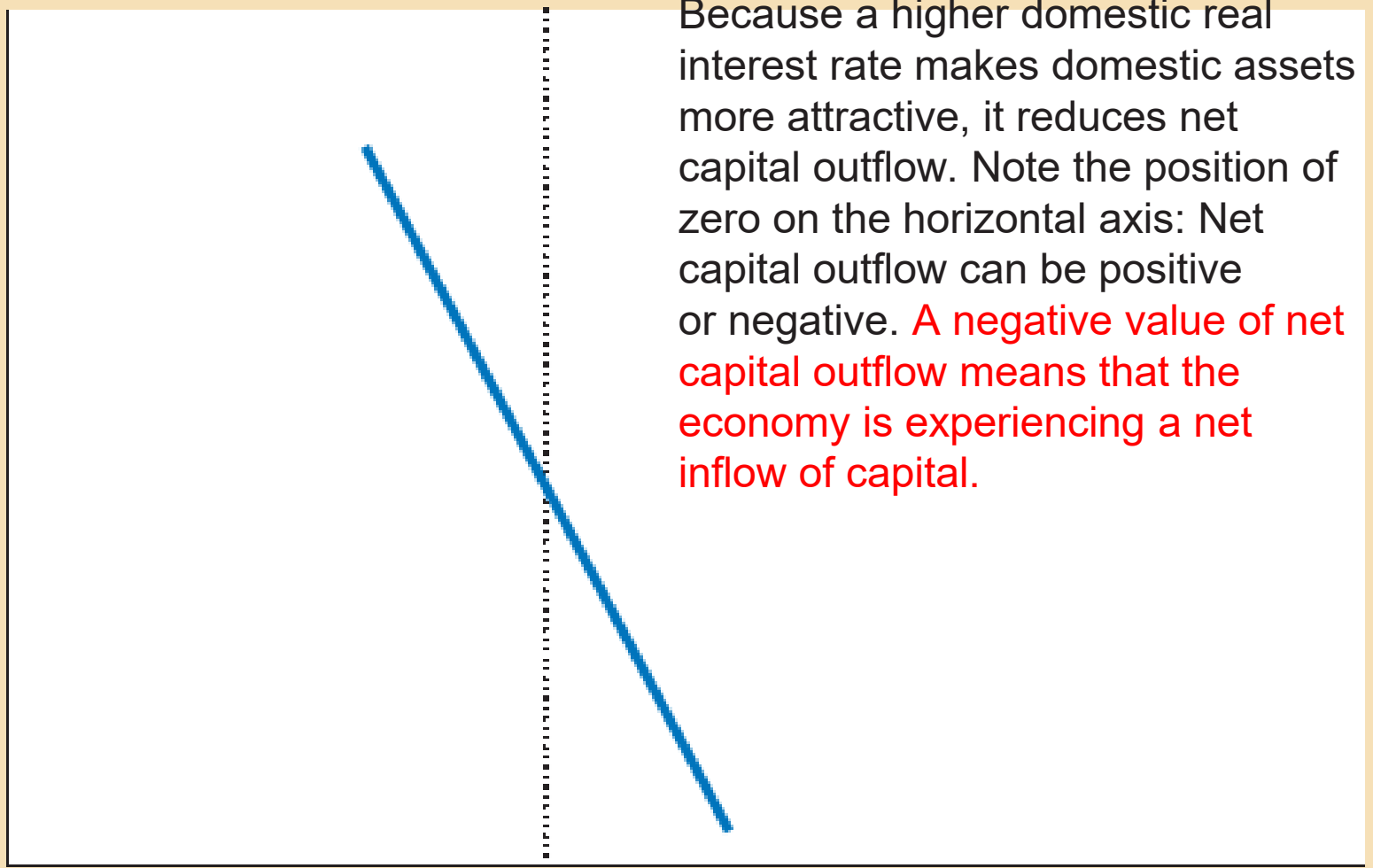
- In the market for loanable funds, supply comes from national saving and demand comes from domestic investment and net capital outflow.
 - In the market for foreign-currency exchange, supply comes from net capital outflow and demand comes from net exports.
- 

The Real Equilibrium in an Open Economy

- Net capital outflow links the loanable funds market and the foreign-currency exchange market.
 - The key determinant of net capital outflow is the real interest rate. **HOW.....?**

How Net Capital Outflow Depends on the Interest Rate

Real
Interest
Rate



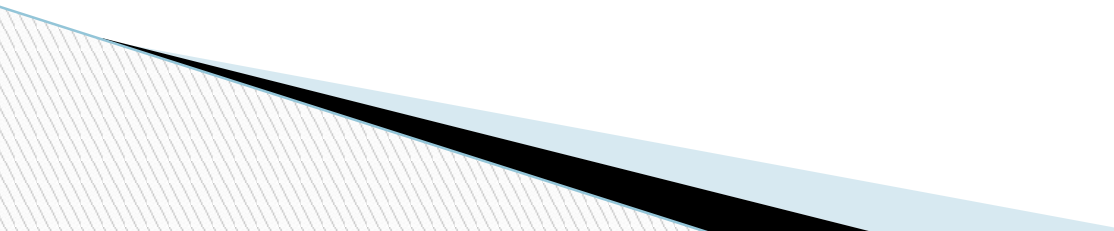
Net capital outflow
is negative.

0

Net capital outflow
is positive.

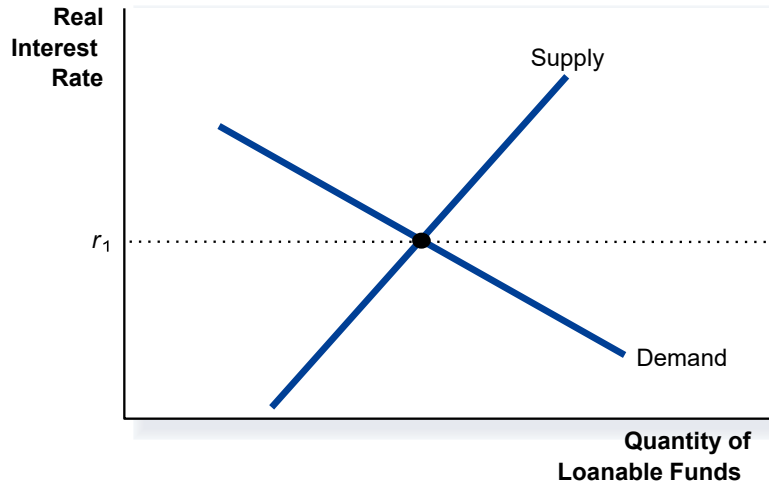
Net Capital
Outflow

The Real Equilibrium in an Open Economy

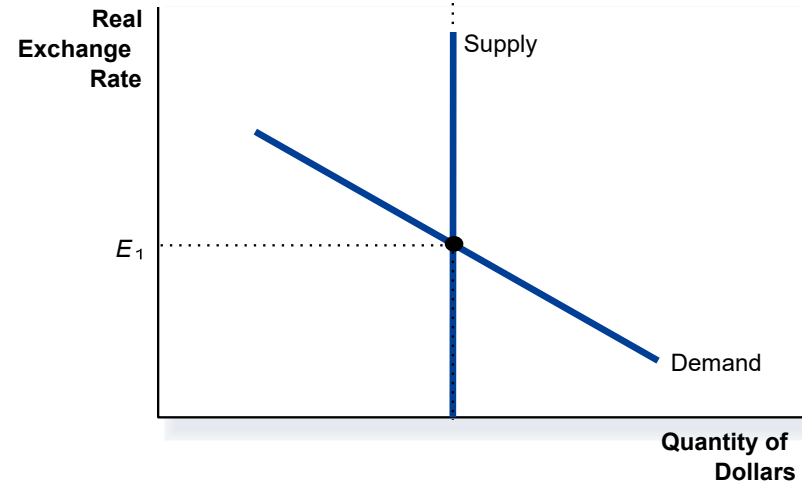
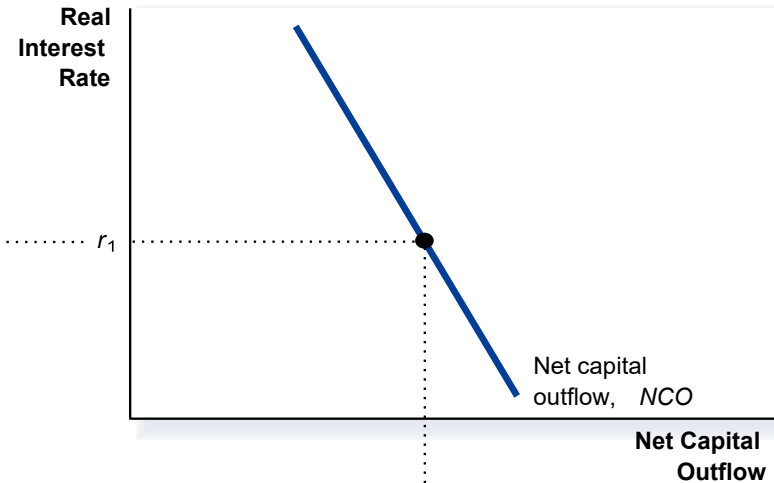
- ❑ Prices in the loanable funds market and the foreign-currency exchange market adjust simultaneously to balance supply and demand in these two markets.
 - ❑ As they do, they determine the macroeconomic variables of national saving, domestic investment, net foreign investment, and net exports.
- 

The Real Equilibrium in an Open Economy

(a) The Market for Loanable Funds



(b) Net Capital Outflow

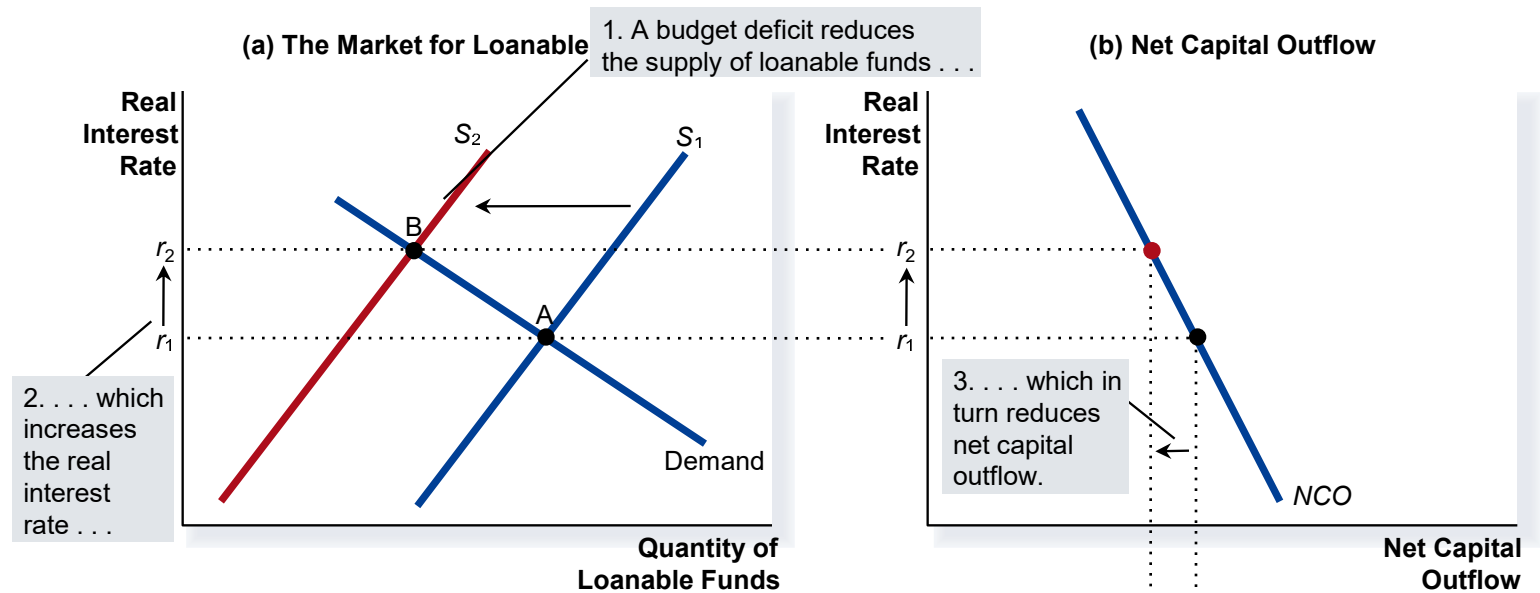


(c) The Market for Foreign-Currency Exchange

How Policies and Events Affect An Open Economy

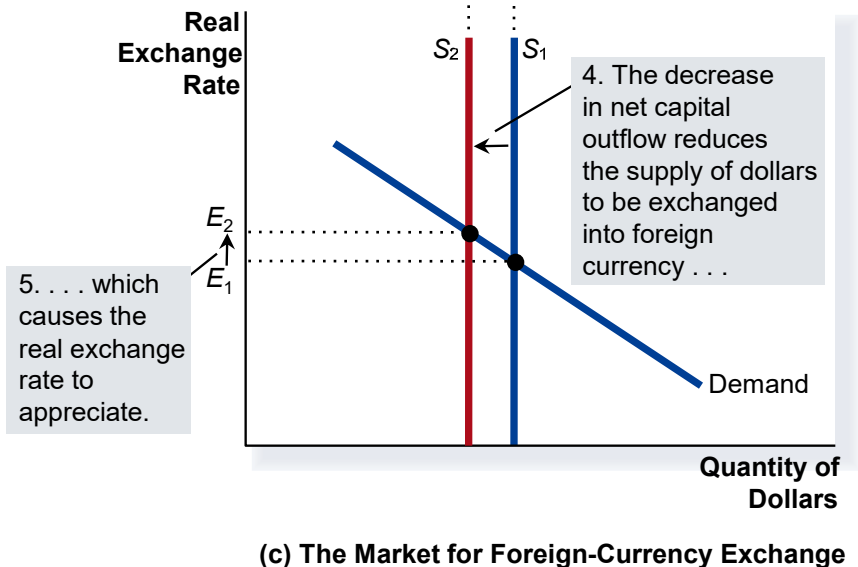
- The magnitude and variation in important macroeconomic variables depend on the following:
 - Government budget deficits
 - Trade policies
 - Political and economic stability

The Effects of Government Budget Deficit



In an open economy, government budget deficits . . .

- reduce the supply of loanable funds,
- drive up the interest rate,
- crowd out domestic investment,
- cause net foreign investment to fall.
- reduce supply of domestic currency
- real exchange rate appreciate



Trade Policy

- A *trade policy* is a government policy that directly influences the quantity of goods and services that a country imports or exports.
 - Tariff: A tax on an imported good.
 - Import quota: A limit on the quantity of a good produced abroad and sold domestically.
- Because they do not change national saving or domestic investment, trade policies do not affect the trade balance.
 - For a given level of national saving and domestic investment, the real exchange rate adjusts to keep the trade balance the same.

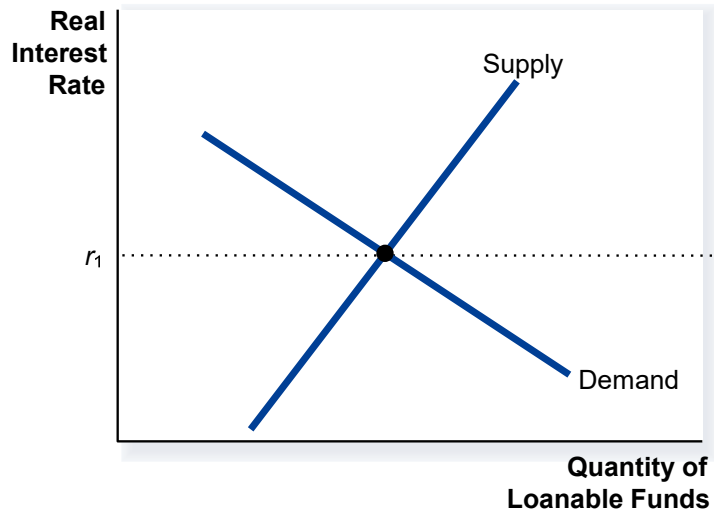
Trade Policy

■ Effect of an Import Quota

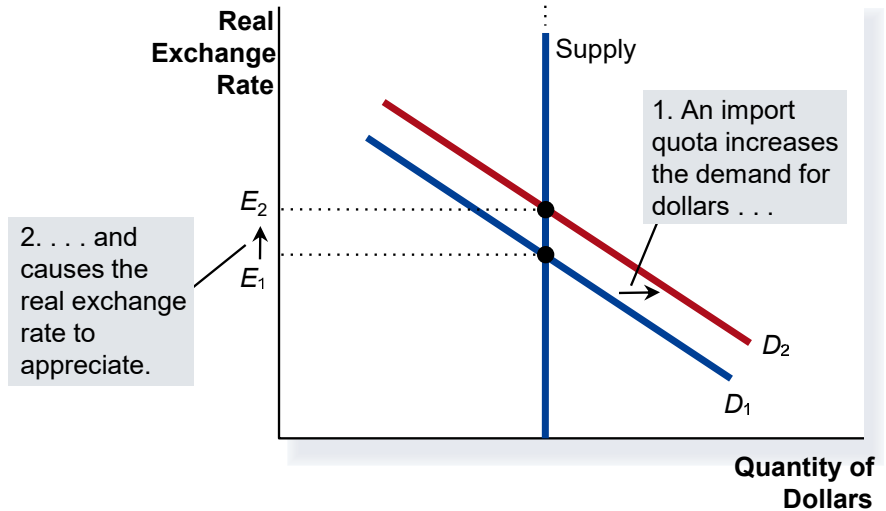
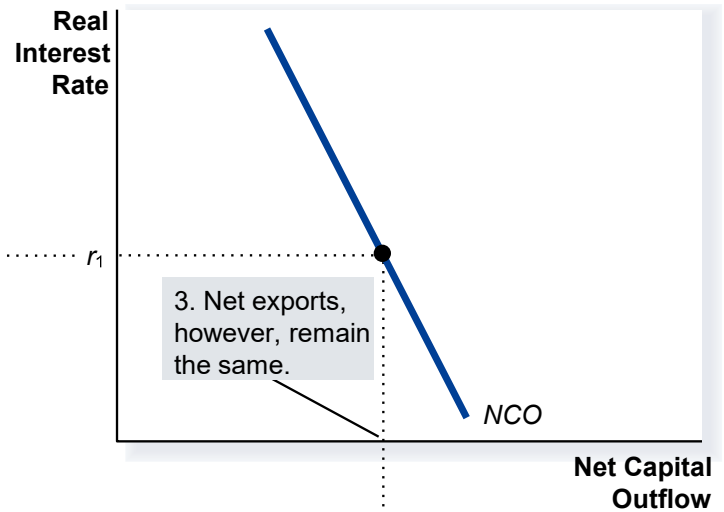
- Because foreigners need dollars to buy U.S. net exports, there is an increased demand for dollars in the market for foreign-currency.
 - This leads to an appreciation of the real exchange rate.
- An appreciation of the dollar in the foreign exchange market encourages imports and discourages exports.
- This offsets the initial increase in net exports due to import quota.

The Effects of an Import Quota

(a) The Market for Loanable Funds



(b) Net Capital Outflow



(c) The Market for Foreign-Currency Exchange

Political Instability and Capital Flight

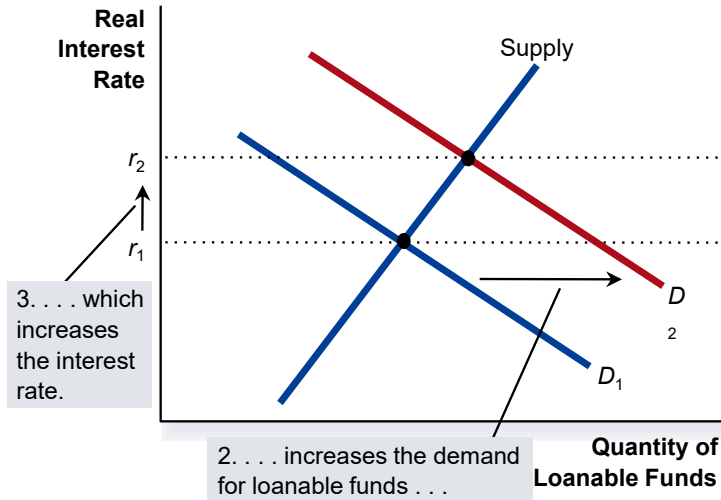
- *Capital flight* is a large and sudden reduction in the demand for assets located in a country.
- Capital flight has its largest impact on the country from which the capital is fleeing, but it also affects other countries.
 - If investors become concerned about the safety of their investments, capital can quickly leave an economy.
 - Interest rates increase and the domestic currency depreciates.

Political Instability and Capital Flight

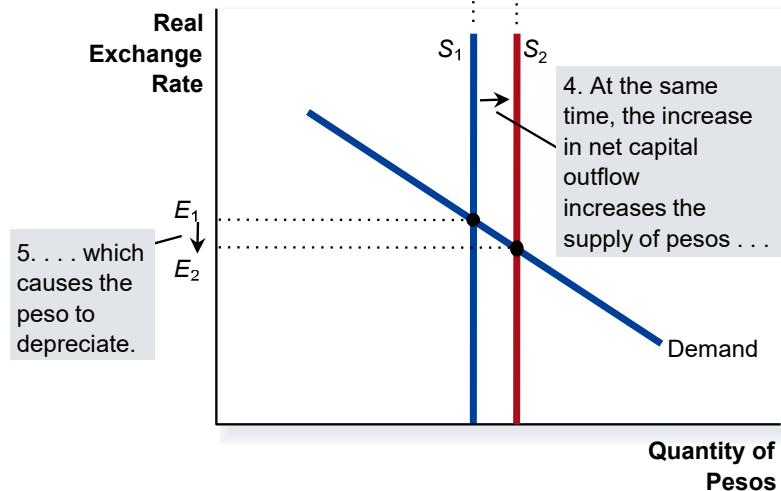
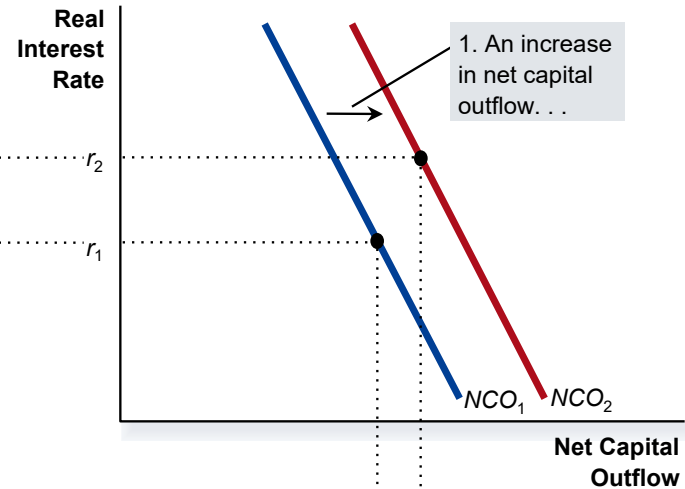
- When investors around the world observed political problems in Mexico in 1994, they sold some of their Mexican assets and used the proceeds to buy assets of other countries.
- This increased Mexican net capital outflow.
 - The demand for loanable funds in the loanable funds market increased, which increased the interest rate.
 - This increased the supply of pesos in the foreign-currency exchange market.

The Effects of Capital Flight

(a) The Market for Loanable Funds in Mexico



(b) Mexican Net Capital Outflow



(c) The Market for Foreign-Currency Exchange

Assignment-7

1. Answer questions for discussion – Samuelson chapter 27 & 28.
2. Answer questions for discussion plus problems and application – Mankiw chapter 18 & 19

THANK YOU